

THE CRAFT OF TEACHING: ENGAGING TECHNIQUES FOR HIGHER EDUCATION



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The Craft of Teaching: Engaging Techniques for Higher Education

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This text was compiled on 04/23/2026

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1: Metacognition

Introduction

Teaching is both an art and a science, requiring faculty to navigate a dynamic landscape of student needs, evolving technology, and ever-changing classroom environments. From addressing common student challenges to designing engaging lectures, instructors must constantly adapt to foster meaningful learning experiences. This textbook serves as a practical guide, offering strategies to enhance student engagement, facilitate productive discussions, and create active learning opportunities in both in-person and online settings.

Whether you're a new instructor or an experienced educator looking to refine your approach, this resource provides insights and best practices to make your teaching more effective and enjoyable. You'll find tools to build inclusive and interactive learning spaces, techniques for keeping students motivated, and methods for leveraging digital platforms to enhance engagement. By incorporating these strategies, you can create a classroom—physical or virtual—where students feel empowered to participate, think critically, and succeed.

Metacognition

Metacognition—thinking about one's own thinking—is a cornerstone of meaningful learning and personal growth. As described in *How Learning Works* (Lovett et al., 2023), metacognition equips students with the tools to plan, monitor, and evaluate their learning. These skills are essential for becoming self-directed learners, capable of adapting strategies to diverse challenges and lifelong learning contexts.

Metacognitive activities can be seamlessly integrated into your course design to complement and strengthen other pedagogical approaches, including active learning, inquiry-based methods, and Universal Design for Learning (UDL). By fostering students' ability to reflect on their learning, metacognition supports engagement, equity, and the development of transferable skills, making it a powerful framework for higher education.

What is Metacognition?

Metacognition involves three interconnected processes:

1. **Planning:** Setting goals, analyzing task demands, and preparing strategies.
2. **Monitoring:** Actively assessing comprehension and progress during learning.
3. **Evaluating:** Reflecting on outcomes to refine future strategies.



Figure 1.1: Metacognitive Cycle

In *How Learning Works*, Lovett et al. (2023) emphasize that metacognition allows students to assess their knowledge and skills, plan their approach, monitor progress, and adjust strategies as needed. This cyclical process transforms students into reflective learners who adapt to challenges and develop lifelong learning habits.

Metacognition also interrelates with other key concepts:

- **Active Learning:** By planning and monitoring their engagement, students take a more active role in their education.
- **Inquiry-Based Learning:** Reflective practices help students generate and refine questions.

- **Feedback:** Metacognition encourages students to actively incorporate feedback into their learning process.
- **Universal Design for Learning:** Supporting metacognitive strategies aligns with UDL's emphasis on providing multiple means of engagement and expression.

Strategies for Fostering Metacognition

Planning for Learning

Planning is the foundation of effective learning. As Lovett et al. (2023) explain, expert learners allocate significant time to planning, while novice learners often overlook its importance. Teaching students how to plan enhances their ability to navigate complex tasks and manage their learning.

Ideas for Implementation:

- Use non-graded pre-tests to help students identify what they already know and what they need to learn.
- Have students preview readings and generate guiding questions before class.
- Model expert planning strategies when solving problems or allocating study time.
- Offer structured study plans as part of specifications grading, allowing students to choose pathways that fit their goals.
- Share examples of effective learning strategies specific to your course.

Monitoring for Learning

Monitoring involves regularly assessing understanding and progress. This process helps students identify gaps in their knowledge and refine their approach during learning tasks.

Ideas for Implementation:

- Assign reflective “minute papers” or “muddiest point” activities during or after class.
- Encourage the use of metacognitive note-taking techniques, like the Cornell system, which integrates space for questions and summaries.
- Have students generate potential test questions as part of self-assessment.
- Use guided reading questions to reinforce comprehension and promote active reading.
- Incorporate confidence judgments during quizzes, encouraging students to reflect on their certainty and performance.

Evaluating Learning

Evaluation closes the metacognitive loop, prompting students to analyze their performance and make adjustments for future tasks. Lovett et al. (2023) emphasize that effective evaluation encourages students to develop resilience and adaptability.

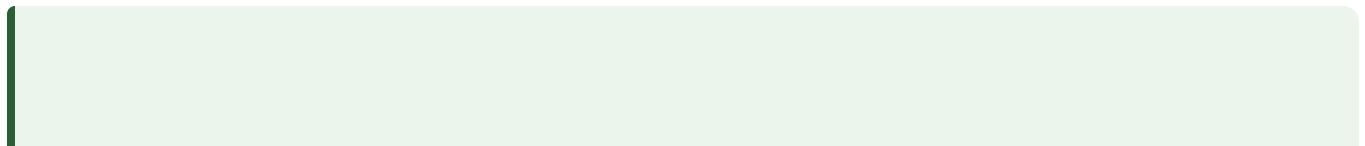
Ideas for Implementation:

- Assign “exam wrappers” that guide students to reflect on their preparation, performance, and strategies for improvement.
- Include reflection components in major assignments, asking students to analyze what worked and what didn't.
- Ask students to formally respond to feedback, identifying areas for revision and growth.
- Require strategy projects where students outline their approach to a task and evaluate its effectiveness

Why Metacognition Matters

Metacognition not only fosters academic success but also prepares students for professional and personal challenges. It aligns with principles of active learning by encouraging students to take ownership of their education, and it supports equity by empowering all learners to develop skills for success, regardless of their starting point.

While many students arrive at college with limited metacognitive skills, these can be taught and cultivated (Cohen, 2012; Steiner, 2016). Evidence shows that embedding metacognitive activities into specific courses, rather than teaching them as stand-alone lessons, enhances transferability and impact (Steiner, Dean, Foote, & Goldfine, 2016).



Key Takeaway

By integrating metacognition into your teaching practices, you provide students with the tools to navigate complex challenges, foster critical thinking, and develop habits for lifelong learning. In turn, students become more engaged, self-aware, and adaptable, supporting their success both in and beyond the classroom.

? Discussion Questions

1. Planning for Learning: What are some specific ways you have encouraged students to take an active role in planning their learning? Have you noticed any differences in student performance when they engage in structured planning activities?
2. Monitoring for Learning: How do you currently help students assess their own understanding during a lesson? What challenges have you encountered in getting students to accurately identify gaps in their knowledge?
3. Evaluating Learning: Reflection is a key component of learning. What strategies have you used to encourage students to analyze their performance and adjust their study habits accordingly?

Sources and Attribution

Primary Source

This section is informed by and adapted from:

- Kennesaw State University, Faculty Development and Recognition. Self-Directed Learners. Available at: [Kennesaw State Faculty Development \(opens in new window\)](#)

For a full list of references and additional resources, please follow the link above.

Use of AI in Section Development

This section was developed using a combination of existing research, expert-informed insights, and AI-assisted drafting. ChatGPT (OpenAI) was used to:

- Summarize key principles of self-directed learning into a structured and accessible guide for educators.
- Clarify best practices for fostering student autonomy, metacognitive skills, and self-regulated learning.
- Enhance readability and coherence, ensuring that strategies for developing self-directed learners are both research-supported and practically applicable.

While AI-assisted drafting provided a structured foundation, all final content was reviewed, refined, and contextualized to ensure accuracy, pedagogical effectiveness, and alignment with cited sources. This section remains grounded in institutional best practices and respects Creative Commons licensing where applicable.

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2: Where Student Challenges Meet Teaching Strategies

Education's goal is to unlock potential, no matter the starting point.

Students in technical colleges come from diverse backgrounds and face a variety of challenges that influence their learning experiences. Unlike traditional academic institutions, technical colleges focus on career-specific training, requiring educators to not only teach theoretical concepts but also develop students' practical, hands-on skills for the workforce. However, various academic and personal obstacles—such as gaps in foundational knowledge, time management struggles, financial barriers, and mental health concerns—can impact student success.

As an educator, getting started with designing your teaching—whether in an online or traditional classroom—requires thoughtful planning to create an inclusive and engaging learning environment. Effective course design should consider student needs, incorporate active learning strategies, and leverage technology to enhance accessibility. For online courses, ensuring clear communication, interactive materials, and structured support is crucial. In traditional classrooms, fostering collaboration, providing hands-on experiences, and offering flexibility can help address diverse learning styles and challenges.

By implementing adaptable teaching strategies, educators can support students in overcoming barriers and developing the skills needed for career success. Let's begin by focusing on our students, the core of our work, and the true measure of our success. The challenge lies in their differences; each student arrives with unique experiences, abilities, goals, and varying levels of support. No two learners are alike, making it essential for educators to recognize and adapt to their individual needs. The real challenge for faculty is finding ways to support all students simultaneously. This is, without a doubt, an ambitious task, but one that is essential for fostering an efficient and effective learning environment.

To meet this complex challenge, teaching strategies must be flexible and responsive, adapting to where students are in their educational journey and what they need most at the moment. Effective teaching strategies may shift fluidly across the elements of the [Guided Pathways Model](#), responding to the evolving needs of students rather than following a rigid sequence. While instructors might focus on ensuring students are learning through active engagement or on helping students get on a path with early career exploration activities, the goal remains the same: supporting student progress. At times, a strategy may contribute to clarifying paths to student education and career goals, while other times it may focus on keeping students on their path through timely feedback and academic support. Regardless of where a strategy fits within the framework at any given moment, the ultimate aim is to foster meaningful learning experiences that lead to successful outcomes in both education and career.

Since students' success is our goal, let us first take a look at what some of the obstacles our students are facing and consider each as we develop our instruction. As you read through each, make a list or mental note of the student struggles you commonly see in your classroom. Add to this list if needed.

1. **Academic Preparedness Gaps** – Some students may struggle with foundational skills in math, reading, or critical thinking, making it difficult to grasp technical concepts.
2. **Time Management and Work-Life Balance** – Many students juggle coursework with jobs, family responsibilities, and other commitments, leading to stress and difficulty meeting deadlines.
3. **Financial Struggles** – Tuition costs, textbooks, and living expenses can be barriers to student retention and focus.
4. **Learning Disabilities and Neurodiversity** – Conditions such as dyslexia, ADHD, and autism spectrum disorders require adaptive teaching strategies and additional support.
5. **Language Barriers** – English Language Learners (ELLs) may struggle with technical vocabulary and communication, affecting comprehension and participation.
6. **Lack of Motivation or Career Uncertainty** – Some students enroll without a clear career goal, leading to disengagement and difficulty staying motivated.
7. **Mental Health Challenges** – Anxiety, depression, and stress can impact concentration, attendance, and academic performance.
8. **Digital Literacy & Limited Access to Technology** – Students come to college with a variety of experiences in technology. Getting students where they need to be to interact with the learning materials, typically provided through digital means, can be a challenge for many students. Additionally, some students may lack reliable internet or up-to-date computers, affecting their ability to complete coursework.

9. **Adapting to Hands-On Learning** – While some students excel in hands-on training, others may struggle with practical application and require additional support.

? Discussion Question

Brainstorm the common struggles you see in your recent experiences.

Recognizing the common challenges our students face is only the first step. We must go further. It is essential to explore how our teaching strategies can help address these obstacles, ensuring that all students have the opportunity to succeed. By thoughtfully aligning our instruction with student needs, we can create a more supportive learning environment that fosters engagement and growth.

Beyond adapting our teaching methods, we must also connect students with available college resources and empower them to advocate for themselves. Providing guidance on academic support services, financial assistance, and mental health resources can make a significant difference in student success. Equipping students with the skills to communicate their needs and seek help when necessary builds confidence and resilience, preparing them not only for the classroom but also for their future careers. However, all of the support at a college cannot provide success if the opportunities for learning are not clear in the classroom.

Addressing Student Challenges Through Effective Teaching Strategies

To foster student success, educators must implement teaching strategies that directly address the challenges students face. Below are some targeted approaches for each issue identified. Add your experiences and methods for tackling the tough issues:

Addressing Academic Preparedness Gaps

Strategy: Implement scaffolding techniques, which involve breaking down complex concepts into manageable steps and providing supplemental resources such as tutoring or online modules. Incorporating real-world examples and hands-on activities helps students apply foundational skills in a practical context.

Your Strategy:

Supporting Time Management and Work-Life Balance

Strategy: Using structured calendars, assignment reminders, and time management workshops that model industry standards can help students stay organized and meet deadlines.

Your Strategy:

Assisting Students with Financial Struggles

Strategy: Utilize cost-effective teaching materials, such as open educational resources (OER), digital textbooks, and free industry tools.

Your Strategy:

Supporting Students with Learning Disabilities and Neurodiversity

Strategy: Apply Universal Design for Learning (UDL) principles by offering multiple means of engagement, representation, and expression. For example, allow students to demonstrate their knowledge through written, verbal, or hands-on assessments.

Your Strategy:

Helping English Language Learners (ELLs)

Strategy: Use visual aids and bilingual text/glossaries to enhance comprehension.

Your Strategy:

Motivating Students and Clarifying Career Goals

Strategy: Integrate industry guest speakers to help students connect coursework to career paths.

Your Strategy:

Addressing Mental Health Challenges

Strategy: Create a supportive classroom environment by normalizing discussions about mental health and promoting available resources, such as counseling services.

Your Strategy:

Bridging the Digital Divide

Strategy: Training students in digital literacy and incorporating step-by-step guides on using online learning platforms ensures equitable access to course materials.

Your Strategy:

Supporting Students Adapting to Hands-On Learning

Strategy: Use a blended approach that combines theoretical instruction with step-by-step demonstrations and guided practice.

Your Strategy:

? Activity: Creating a New Teaching Strategy

Objective: Develop targeted teaching strategies by brainstorming about a student with support needs from the list of common challenges in technical colleges.

Instructions:

1. **Choose a Student Profile:** *Select one* of the student challenges from the list discussed in the chapter that you commonly see in your classrooms. For example:
 - A working parent struggling with time management
 - A student with digital literacy and access issues
 - A student with a learning disability needs alternative learning methods
2. **Analyze the Student's Needs:** Consider the unique challenges this student might face in your course. Ask yourself:
 - Describe what fills their day-to-day schedule from sun-up to sun-down.
 - What barriers might prevent them from succeeding?
 - How might their background or circumstances impact engagement and learning?
 - What strengths or prior experiences could they bring to the classroom?
3. **Brainstorm Teaching Strategies:** In small groups or individually, outline at least three strategies to support this student. Knowing what you know about their day, their barriers, and their experiences, think about:
 - How course design changes could enable students to learn more effectively.
 - Consider specific classroom supports that could bridge the gaps for students.
 - Rethink how your time with students can best be spent to enhance learning, practice skills, and develop confidence in the industry. (Be specific.)
4. **Share and Discuss:** Present your plan to the group and discuss different approaches. Compare ideas and consider additional ways to make the classroom more inclusive for all students.

Reflection Questions

- How did this activity challenge your assumptions about student needs?
- What new strategies did you learn from others that you could apply in your own teaching?
- How can these strategies be adapted to support multiple types of students?

By engaging in this activity, educators can proactively design inclusive teaching strategies that help all students—regardless of background or challenges—thrive in technical college courses.

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3: Course Design for Engagement and Opportunity

Good course design focuses on what students need, as well as what you want to teach.

One of the most effective ways to ensure that students thrive in their academic journey is through engaging course design and the provision of diverse learning opportunities. A well-designed course fosters motivation, deepens understanding, and improves student retention, making education more meaningful and impactful. Engagement is a key factor in student success. When students find a course stimulating and interactive, they are more likely to invest time and effort into their learning. An engaging course design incorporates elements that captivate students' interests, promote active participation, and create a dynamic educational experience. This can be achieved through interactive discussions, hands-on activities, and real-world applications that connect course content to students' lives. While many faculty members may note that they do not develop their own courses or must follow a prescribed curriculum, effective course design goes beyond just the content. As you will see, the way material is delivered can be just as impactful as the curriculum itself in creating an engaging learning experience.

Benefits of Engaging Course Design

1. **Enhances Student Motivation:** Courses that are designed with engaging content encourage students to take an active role in their learning. Incorporating diverse teaching methods such as multimedia, gamification, and collaborative projects keeps students motivated and eager to participate.
2. **Promotes Active Learning:** Active learning strategies, such as case studies, problem-solving exercises, and group discussions, allow students to apply theoretical concepts in practical ways. This approach leads to deeper comprehension and long-term retention of information.
3. **Encourages Critical Thinking and Creativity:** When students are challenged to analyze, evaluate, and create, they develop critical thinking and problem-solving skills. Engaging course design fosters intellectual curiosity and encourages students to explore new ideas beyond textbook knowledge.
4. **Improves Retention and Academic Performance:** Research shows that students who are actively engaged in their courses are more likely to stay enrolled and perform better academically. Engaging course design minimizes passive learning, reducing dropout rates and increasing student success.
5. **Supports Diverse Learning Needs:** Students come from varied backgrounds and have different learning styles. Offering multiple means of instruction—such as visual, auditory, and kinesthetic learning approaches—ensures that all students have equal opportunities to grasp course content.

Lecturing as Active Learning

Lectures remain a cornerstone of teaching in higher education, but when delivered as passive monologues, they often fall short of engaging students or fostering deep learning. To transform lecturing into a High-Impact Teaching Practice (HITP), instructors can incorporate active learning strategies tailored to their modality—whether online, in a large lecture hall, or in a small seminar setting. These strategies promote student engagement, critical thinking, and retention, aligning lecturing with the principles of backward design, Universal Design for Learning (UDL), and collaborative learning.

Key Strategies for Active Learning

Chunk Content for Better Retention

- Break your lecture into smaller, digestible segments of 10–15 minutes.
- After each segment, engage students with activities, reflections, or questions.
- In large lectures, tools like clickers or polling apps can gather responses efficiently. Online, use breakout rooms or chat for small-group discussions.

✓ Example

In an online biology lecture, present the stages of mitosis in 10-minute chunks, followed by a poll where students match each stage to a visual diagram.

Scaffolding for Deeper Understanding

- Begin with foundational concepts and build progressively toward more complex ideas.
- Use analogies, visuals, and real-world examples to bridge knowledge gaps.
- Provide scaffolding in the form of guided notes or outlines, which students can complete during the lecture.

✓ Example

In a small seminar on literature, start by discussing the historical context of a novel, then scaffold toward analyzing a specific passage for thematic elements. Use guided prompts to direct the analysis.

Pause for Reflection and Discussion

- Incorporate short pauses every 12–18 minutes to allow students to process information, rework notes, or discuss a question.
- The Pause Procedure is especially effective in large classes, giving students time to reflect and engage without losing momentum.

✓ Example

In a large lecture hall, pause to ask students to write down one question they have about the material and discuss it with a neighbor. Online, use a chat prompt like “What’s one takeaway from this section?”

Integrate Mini-Quizzes and Polls

- Use low-stakes quizzes or polls to reinforce learning and check comprehension.
- Tools like Poll Everywhere, Mentimeter, or Kahoot are effective for in-person and online lectures.
- In small seminars, verbal quizzes or quick individual responses can achieve the same goal.

✓ Example

During an economics lecture, use a poll to ask students which economic model applies to a hypothetical scenario. Follow up by explaining the correct answer and its rationale.

Use Demonstrations and Predictions

- Engage students by having them predict the outcome of a demonstration or scenario before revealing the result.
- Follow up with a discussion of why the prediction was correct or incorrect.

✓ Example

In a physics class, ask students to predict how changing a variable in an experiment will affect the outcome. Conduct the experiment live or through a video and discuss the results.

Build in Opportunities for Reflection

- Incorporate “Minute Papers,” where students write briefly about what they’ve learned or what remains unclear.
- Encourage students to connect lecture content to their prior knowledge or real-world applications.

✓ Example

In a public health lecture, ask students to write for one minute on how the day’s topic applies to a current health issue in their community. Online, use a shared Google Doc for responses.

Adapting to Different Modalities

Large Classes

- Use technology to maintain engagement: polling tools, live quizzes, or digital Q&A platforms like Slido.
- Encourage peer learning with structured think-pair-share activities.
- Use visuals and multimedia to break up the lecture and maintain interest.

Small Seminars

- Foster discussion by posing open-ended questions and allowing students to take the lead in parts of the lecture.
- Use the seminar format to dive deeper into analysis or application, supported by reflective activities.

Online Lectures

- Leverage breakout rooms for small-group discussions and collaborative problem-solving.
- Use asynchronous tools like discussion boards for reflection and interaction between live sessions.
- Provide clear instructions for active learning activities and ensure the accessibility of all materials.

Lecturing as a High-Impact Teaching Practice (HITP)

By integrating active learning into lectures, instructors transform a traditionally passive format into an engaging, collaborative experience that embodies HITPs. This approach:

- Encourages metacognition by prompting students to reflect on their learning.
- Fosters collaborative learning through peer discussions and shared activities.
- Aligns with learning-centered course design by focusing on significant learning outcomes rather than content coverage.
- Builds student confidence by demonstrating that their participation and feedback shape the learning process.

Example Lecture Flow

1. **Introduction (5 minutes):** Present the lecture's goals and key questions. Use a quick poll to assess students' prior knowledge.
2. **First Chunk (10–15 minutes):** Explain a key concept with visuals and examples. Use scaffolding to connect new ideas to prior material.
3. **Pause and Activity (5 minutes):** Ask students to discuss the concept with a partner or solve a related problem individually. Share responses as a class.
4. **Second Chunk (10–15 minutes):** Introduce a second concept, building on the first. Use a mini-quiz to check for understanding.
5. **Reflection (5 minutes):** Use a "Muddiest Point" prompt: "What was the most confusing part of today's lecture?"
6. **Conclusion (5 minutes):** Summarize key points and connect them to upcoming topics or assignments.

Practical Tips for Instructors

1. **Record and Reflect:** Record your lecture to evaluate pacing, clarity, and engagement strategies.
2. **Plan Timing:** Use a timer to ensure you balance content delivery with active learning opportunities.
3. **Gather Feedback:** Use student feedback to refine your methods and identify what works best in different contexts.
4. **Start Small:** If you're new to active lecturing, begin by incorporating one or two strategies, such as pauses for reflection or mini-quizzes, and build from there.

By embracing these principles, instructors can create lectures that are not only informative but also engaging and impactful, ensuring that students leave with a deeper understanding and lasting skills.

Facilitating Discussions

Discussions can be a powerful teaching strategy, enhancing student motivation, intellectual agility, and democratic habits. When thoughtfully planned and facilitated, they provide a space for students to articulate and defend positions, evaluate evidence, and consider multiple perspectives. However, discussions are inherently unpredictable and require instructors to surrender some control over the flow of information. This unpredictability can be daunting, but with careful preparation, discussions can achieve a balance between lively exploration and focused learning.

This section explores strategies to plan and lead effective discussions, organized around three critical aspects: **cognitive**, **social/emotional**, and **physical factors**.

Cognitive Factors: Guiding the Intellectual Process

Determine and Communicate Learning Objectives

Effective discussions begin with a clear purpose. What should students learn or practice during the discussion? Your objectives might include:

- **Analyzing Arguments:** Compare evidence and perspectives in assigned readings.
- **Creative Problem-Solving:** Develop and defend a design for a sustainable city.
- **Practical Application:** Formulate policy recommendations for a contemporary issue.

Clearly articulating and sharing these objectives with students helps focus their thinking and motivates meaningful participation.

Plan a Strategy

A well-thought-out plan provides structure while allowing room for flexibility. Key considerations include:

- **Preparation:** Should students read a case study, complete a reflection, or watch a video beforehand?
- **Framing Questions:** What questions will stimulate discussion and deeper analysis?
- **Participation Formats:** Will the discussion involve the whole class, small groups, or pairs?
- **Managing Challenges:** How will you address dominance, digressions, or silence?

Having a plan ensures that discussions align with learning goals and adapt to students' needs.

Ask Good Questions

Good questions spark engagement and promote critical thinking. Consider these types:

- **Exploratory:** Probe for facts or foundational knowledge.
- **Relational:** Draw comparisons between ideas or themes.
- **Hypothetical:** Explore the implications of changes in context.
- **Summary:** Synthesize key points and insights.

Start with convergent questions (e.g., "What are the main arguments in the reading?") to establish shared understanding, then transition to divergent questions (e.g., "What are the broader implications of these arguments?") for deeper exploration.

Common Questioning Errors

When instructors are nervous that a discussion might flag, they tend to fall prey to some common questioning errors. These include:

Asking too many questions at once: Instructors often make the mistake of asking a string of questions together, e.g., "What do you think the author is trying to say here? Do you agree with him? Is his evidence convincing? Did you like this article?" Students may get confused trying to figure out which question to address first. Asking a number of questions together may also conflate issues you really want to help students distinguish (for example, the author's thesis versus the kinds of evidence he uses to support it).

Asking a question and answering it yourself: We have all had the experience of asking a question only to encounter blank stares and silence. The temptation under these circumstances is to jump in and answer your own question, if only to relieve the uncomfortable silence. Don't assume, though, that students' silence necessarily indicates that they are stumped (or unprepared); sometimes, they are simply thinking the question through and formulating an answer. Be careful not to preempt this process by jumping in too early.

Failing to probe or explore the implications of answers: One mistake instructors can make in leading a discussion is not to follow up sufficiently on student contributions. It is important not only to get students talking but to probe them about their

reasoning, ask for evidence, explore the implications of what they say, etc. Follow-up questions push students to think more deeply, to substantiate their claims, and to consider the practical impact of particular perspectives.

Asking unconnected questions: In the best discussions, there is a logical progression from question to question so that, ultimately, the discussion tells (or reveals) a story. When you are planning your discussion questions, think about how they fit together.

Asking yes/no or leading questions: Asking questions with a yes/no answer can be the starting point of a good discussion, but only if there is a follow-up question that calls for explanation or substantiation. Otherwise, yes/no questions tend to be conversation-stoppers. By the same token, discussions can stall if the instructor's questions are overly leading, i.e., if there is clearly an answer the instructor wants, and the students' task is simply to guess it, rather than to think for themselves.

Ignoring or failing to build on answers: If students do not feel like their voices have weight in the discussion, their motivation to participate drops. Thus, it is important to acknowledge student contributions, responding enthusiastically when they are insightful ("That's an excellent point, Sarah; could you elaborate further?") and pointing out when they contain inaccuracies or problematic reasoning ("Take another look at the article, Trinh; is that really what the author is claiming?"). If you do not wish to play such a directive role yourself – and want students to develop the habit of assessing and responding to one another's contributions – you can throw student comments back to the class for evaluation (for example, "Do the rest of you agree with John's recommendation? What would be some possible consequences if this plan of action were followed?")

Provide Direction and Maintain Focus

Discussions thrive with a clear focus. Use strategies like summarizing key points mid-discussion or redirecting when off-track (e.g., "How does this connect to our main question?"). Recognize signs of stagnation, such as repetition or apathy, and reinvigorate with new questions or group formats.

Bring Closure

Leave time to synthesize the discussion:

- Summarize key insights and unresolved questions.
- Assign a reflection or ask students to identify the most important takeaway.
- Provide 2-3 "big picture" conclusions to reinforce learning objectives.

Social and Emotional Factors: Building Trust and Engagement

Demonstrate Relevance

Explain how discussion skills—such as articulating positions or debating respectfully—translate to academic and professional success. Connect topics to students' interests, contemporary issues, or future goals (e.g., "Think about how you'd explain this policy in a job interview.").

Encourage Participation

Create a supportive environment by:

- **Setting Expectations Early:** Foster participation from the first class with icebreakers and ground rules (e.g., "Speak respectfully and avoid put-downs").
- **Ensuring Preparation:** Assign pre-discussion work to focus student thinking (e.g., reflection prompts, discussion questions).
- **Modeling Good Behavior:** Demonstrate active listening, respectful disagreement, and thoughtful engagement.
- **Providing Alternatives:** Use discussion boards or small-group work for students hesitant to speak in large groups.

Monitor Group Dynamics

Balance participation by encouraging quieter students and managing dominant voices. Use strategies like calling on students equitably or asking dominant contributors to reflect while others speak.

Anticipate and address emotional or sensitive moments with strategies like pausing for written reflections or reinforcing ground rules.

Make High-Quality Participation Count

Define and reward meaningful contributions through a rubric. For example, high-quality participation might include synthesizing ideas, providing evidence, or respectfully challenging perspectives.



Figure 3.1: Implementing High-Impact Teaching Practices

Physical Factors: Creating a Conducive Environment

1. **Arrange the Classroom:** Seating arrangements should align with the discussion goals. Use a circle or U-shape for dialogic engagement or clusters for small-group work. For larger classes, use strategies like polling or rotating spokespeople to ensure participation.
2. **Adapt to Constraints:** Even in less-than-ideal spaces, discussions can thrive. For example, in lecture halls, use tools like microphones or discussion boards to facilitate interaction. Use visuals, such as whiteboards or slides, to track progress and focus attention.

Effective discussions require thoughtful attention to cognitive, social/emotional, and physical dimensions. By setting clear goals, asking good questions, fostering an inclusive environment, and creating a supportive physical space, instructors can transform discussions into vibrant learning experiences that engage students deeply and develop critical skills.

Thinking Routines from Project Zero

Thinking routines are simple, flexible activities designed to scaffold students' thinking and make their cognitive processes more visible. Originally developed by researchers at Harvard's Project Zero, these routines support thoughtful engagement with content across disciplines, age groups, and learning environments.

Each routine consists of a small set of questions or steps that help students organize, extend, or reflect on their thinking. They can be used at any point in a lesson—whether to introduce a topic, explore ideas in depth, or reflect on learning. By encouraging learners to recognize and name their “thinking moves,” these routines promote metacognition, curiosity, and deeper understanding.

A wide range of thinking routines and suggestions for how to apply them can be found on the [Project Zero website \(opens in new window\)](#). We list and summarize a few of the core thinking routines below.

Core Thinking Routines

These are simple routines that are applicable across disciplines, topics, and age groups and can be used at multiple points throughout a learning experience or unit of study. These are a good place to start if you or your students are new to thinking routines. Find more Thinking Routines and applications at the [Project Zero website \(opens in new window\)](#). (Note that, as currently designed, you will need to click through the “Resource Links” at the bottom right to access the full PDF of the thinking routine.)

? Circle of Viewpoints

1. Brainstorm a list of different perspectives.
2. Choose one perspective to explore, using these sentence-starters:
 - **I am thinking of ...** the topic ... **from the viewpoint of ...** *the viewpoint you've chosen*
 - **I think ...** *describe the topic from your viewpoint. Be an actor—take on the character of your viewpoint*
 - **A question I have from this viewpoint is ...** *ask a question from this viewpoint*

Purpose: This routine helps students see and explore multiple perspectives. It helps them understand that different people can have different kinds of connections to the same thing and that these different connections influence what people see and think.

Application: The routine works well with topics and artworks that deal with complex issues. It also works well when students are having a hard time seeing other perspectives or when things seem like there are only two sides to an issue. The routine can be used to open discussions about dilemmas and other controversial issues.

? Claim, Support, Question

Drawing on your investigation, experience, prior knowledge, or reading:

- Make a claim about (or give an explanation for, or offer an interpretation of) the topic.
- Identify support (things you see, feel, know) for your claim.
- Ask a question related to your claim or the support. What isn't explained?

Purpose: This routine supports reasoning by having students form claims, explanations, or interpretations and support them with evidence. Questioning claims helps students see reasoning as an ongoing process.

Application: Statements of fact or belief are presented everywhere. This routine is designed to help students take notice of claims and thoughtfully examine them. Use the routine with any topic, work of art, piece of text, poem, etc., to invite careful reasoning and evaluation.

? Compass Points

E = Excited

What excites you about this idea or proposition? What's the upside?

W = Worrisome

What do you find worrisome about this idea or proposition? What's the downside?

N = Need to Know

What else do you need to know or find out about this idea or proposition? What additional information would help you to evaluate things?

S = Stance or Suggestion for Moving Forward

What is your current stance or opinion on the idea or proposition? How might you move forward in your evaluation of this idea or proposition?

Purpose: This routine helps students flesh out an idea or proposition and eventually evaluate it.

Application: This routine works well to explore various sides and facets of a proposition or idea prior to taking a stand or expressing an opinion on it. For instance, the school may be considering the idea of a dress code, a teacher might present the class with the idea of altering the room arrangement, a character in a book might be confronted with making a choice, a politician might be putting forth a new way of structuring taxes, and so on.

? Connect, Extend, Challenge

Consider what you have just read, seen, or heard, and then ask yourself:

- How are the ideas and information connected to what you already know?
- What new ideas did you get that broadened your thinking or extended it in different directions?
- What challenges or puzzles emerge for you?

Purpose: This routine helps students connect new ideas to those they know and encourages them to reflect upon how they have extended their thinking as a result of what they are learning about or experiencing.

Application: Use this routine when you want students to make explicit connections to something previously learned or experienced. Since it is designed to help students process new information actively, it works well as the conclusion to lessons in which students have been reading, watching videos, or otherwise taking in new information. Another approach is to use the routine to close the discussion of a topic or unit of study in order to help students synthesize the information. Some questions you might consider in your planning: Are there connections to be made between this content and what students already know? Will students be engaging with new information that they might find challenging?

? I Used to Think...Now I Think...

Think about what you have learned about the topic we have been studying, and complete the following sentence stems:

- I used to think...
- Now I think...

Purpose: This routine helps students to reflect on their thinking about a topic or issue and explore how and why their thinking has changed. It helps consolidate new learning.

Application: This routine can be used whenever students' initial thoughts, opinions, or beliefs are likely to have changed as a result of a learning experience. This may be after reading new information, watching a film, listening to a speaker, experiencing something new, having a class discussion, at the end of a unit of study, and so on.

? See, Think, Wonder

What do you see?

What do you think about that?

What does it make you wonder?

Purpose: This routine encourages students to make careful observations and thoughtful interpretations. It helps stimulate curiosity and sets the stage for inquiry.

Application: Use this routine when you want students to think carefully about why something looks the way it does or is the way it is. Use the routine with a relevant object (such as an artwork, image, artifact, chart, video, etc.) at the beginning of a new unit to motivate student interest, or try it with an object that connects to a topic during the unit of study. Consider using the routine with an interesting object near the end of a unit to encourage students to apply their knowledge and ideas.

? Think, Pair, Share

Pose a question to students. Give students a few minutes to **think**.

Invite students to **pair** with a nearby student to **share** their thoughts.

Purpose: This routine promotes understanding through active reasoning and explanation. Because students listen to and share ideas with others, it also encourages students to understand multiple perspectives.

Application: This routine can be used when it would help for students to process their thinking aloud with another student. For example, you may ask students to Think, Pair, Share before starting a science experiment, in the middle of solving a math problem, after reading a passage of a book, etc. Sharing can also be done in small groups.

? Think, Puzzle, Explore

What do you think you know about this topic?

What questions or puzzles do you have about this topic?

How might you explore your puzzles about this topic?

Purpose: This routine activates prior knowledge, generates ideas and curiosity, and prepares students for deeper inquiry.

Application: This routine works especially well when introducing a new topic, concept, or theme in the classroom. It helps students recall what they already think they know about the topic and then invites them to identify puzzling questions or areas of interest to pursue. This routine can help you gauge students' current understanding of a topic and inform your subsequent lesson planning. You might return to this thinking routine throughout the study of a topic to surface changes in student conceptions.

? What Makes You Say That?

What's going on?

What do you see that makes you say that?

Purpose: This routine cultivates observation, description, explanation-building, and evidence-based reasoning. Because students share their interpretations, they are encouraged to see multiple perspectives.

Application: Use this routine when you want students to look closely at something and uncover their reasoning about the way it works, how it came to be, or why it is the way it is. For example, you can use this routine when students are looking at works of art or historical artifacts, reading poetry, making scientific observations and hypotheses, or delving into broad conceptual topics like culture, change, or creativity. It is also useful for gathering information on students' prior understandings when introducing a new topic.

3: Course Design for Engagement and Opportunity is shared under a [CC BY-NC-SA 4.0](#) license and was authored, remixed, and/or curated by Philip Smith.

4: Engaging Minds with Active Learning

The best ideas come to those who engage.

Active learning is a teaching approach that encourages students to participate actively in their education through discussion, problem-solving, and collaboration. Rather than passively listening to lectures, students engage with the material in ways that require them to think critically and apply their knowledge. This hands-on approach allows them to explore concepts more deeply and develop a stronger grasp of the subject matter.

By using techniques like group discussions, case studies, and interactive exercises, active learning helps students move beyond memorization and into meaningful engagement. These methods encourage them to analyze information, make connections between ideas, and develop problem-solving skills. This process not only enhances understanding but also makes learning more dynamic and enjoyable, helping students retain information more effectively.

One of the key benefits of active learning is its adaptability to different learning styles and subjects. Educators can experiment with various techniques to find what works best for their students and course content. Whether through hands-on projects, peer collaboration, or guided inquiry, this flexible approach ensures that learning remains engaging and impactful. By tailoring methods to fit different needs, students can develop a deeper appreciation for the material and gain skills that extend beyond the classroom.

Common Active Learning Techniques

While the active learning strategies described in this section vary in form and complexity, they share several **core characteristics**:

- **Student-centered:** Each technique shifts the focus from instructor delivery to student engagement and participation.
- **Collaborative or reflective:** Most involve interaction with peers or structured self-reflection.
- **Constructivist:** Students actively build understanding through doing—generating, applying, discussing, organizing, or evaluating ideas.
- **Flexible:** Many of these strategies can be adapted for use in small or large classes, and in face-to-face, hybrid, or online formats.

At the same time, these strategies differ in how they are **best applied**, depending on your goals:

- **To surface opinions or spark discussion:** Try Opinion Line-Up, Polling and Voting, or Structured Debates. These help students clarify their perspectives and engage with diverse views.
- **To organize or deepen ideas:** Try Sticky-Note Clustering, Concept Mapping, or Cumulative Brainstorming to foster synthesis, organization, and integrative thinking.
- **To apply or transfer knowledge:** Try Application Cards, Decision-Making Scenarios, or Case-Based Decision-Making to promote real-world relevance and higher-order thinking.
- **To encourage peer engagement and feedback:** Peer Review, Think-Pair-Share, and Fishbowl Discussions offer structured interaction and formative learning moments.
- **To prompt individual reflection and quick feedback:** Minute Reflections and Polling provide low-barrier ways to check understanding and support metacognition.

By choosing strategies that align with your learning outcomes, classroom context, and students' needs, you can make active learning a consistent and purposeful part of your teaching practice.

✓ Opinion Line-Up

Opinion Line-Up is an interactive activity that prompts students to consider their position on a complex issue and discuss it with peers. By physically or virtually placing themselves along a spectrum of opinions, students clarify their reasoning and engage with diverse perspectives.

✓ Sticky-Note Clustering

Sticky-Note Clustering is a collaborative activity where students generate ideas individually and then work together to categorize and organize them. This activity combines brainstorming with critical thinking to deepen understanding of a topic.

✓ Dotmocracy

Dotmocracy is a visual and participatory decision-making process where students allocate “votes” to express their preferences or priorities. This strategy fosters discussion and collaboration by making group opinions visible.

✓ Fishbowl Discussion

Fishbowl Discussions involve a small group of students engaging in a focused activity or dialogue while the rest of the class observes. This strategy promotes deep engagement, critical reflection, and collaborative learning.

✓ Cumulative Brainstorming

Cumulative Brainstorming encourages students to generate, critique, and build upon ideas collaboratively, fostering critical thinking and teamwork.

✓ Think-Pair-Share

Think-Pair-Share is a collaborative learning strategy that encourages individual reflection, peer discussion, and class-wide sharing to promote deeper understanding and engagement.

✓ Minute Reflections

Minute Reflections are quick, written responses that encourage students to summarize, question, or reflect on their learning. This activity is simple to implement and provides immediate insight into students’ understanding and concerns.

✓ Structured Debates

Structured Debates are formal discussions where students take opposing sides on an issue, defending their positions with evidence and reasoning. This activity develops critical thinking, argumentation skills, and an appreciation for diverse perspectives.

✓ Concept Mapping

Concept Mapping involves creating visual representations of relationships between concepts, fostering understanding of how ideas connect. Students map key terms, ideas, or processes using diagrams, linking them with lines and labels to show relationships.

✓ Application Cards

Application Cards challenge students to apply a concept or skill to a new scenario, fostering transfer of knowledge and critical thinking. By engaging students in this process, instructors can assess students’ ability to make connections and use concepts in novel contexts.

✓ Peer Review

Peer Review involves students exchanging their work and providing feedback to each other, fostering collaboration, critical thinking, and self-assessment skills. By evaluating their peers’ work, students gain insights into their own understanding and

approaches.

✓ Polling and Voting

Polling and Voting involve asking students to respond to questions, scenarios, or problems to gauge understanding, elicit opinions, or spark discussion. It can range from quick polls to more elaborate voting activities that shape the direction of a class discussion.

✓ Decision-Making Scenarios

Decision-Making Scenarios challenge students to take on the role of decision-makers, analyze a complex issue, and propose solutions. This strategy helps students apply knowledge to real-world or hypothetical situations, developing critical thinking and collaborative skills.

✓ Case-Based Decision-Making

Case-Based Decision-Making combines real-world scenarios with the critical thinking process of making and justifying decisions. It engages students in applying theoretical knowledge to practical problems while evaluating multiple perspectives and possible outcomes.

Laying the Foundations for Group Work

What is the idea?

This short chapter discusses a group activity designed to promote team cohesiveness and open-mindedness in the context of a Business Research Methods undergraduate course. Students had to work in groups (of 3 or 4) for the duration of the course to produce a research report addressing an identified business problem, but many groups faced challenges due to the absence of a shared vision and teamwork skills. Similar problems were encountered in another course on Integrated Marketing Communications, where students worked on a lengthy group project. To address these issues, a formative hourly activity was implemented early on in the term. This activity works best with classes of 15-20 students, and it can be implemented in most courses adopting a team-based active learning approach.

Why this idea?

Team-based active learning is based on the cooperative learning pedagogy and produces numerous benefits for individual learners (Espey 2018). Permanent student groups (ie, groups that are formed for the duration of the course), unlike temporary ones, require students to tackle interpersonal difficulties and to appreciate the benefits of group work (Michaelsen 1992). Team processing skills, one of the elements of cooperative learning, are conducive to building a learning orientation, yet for students undertaking lengthy group assignments, team cohesiveness needs to be built from the onset (Laverie et al., 2008).

The activity was implemented in 5 different small cohorts within two different undergraduate courses, in both of which students had to undertake a demanding group project. The activity was inspired by the need to address practical challenges encountered in class and by other activities discussed in Dawson (2016) and Parker and Goldrick-Jones (2011).

In the first part of this activity (role-play), students work individually, drawing freely on their own experiences and understanding of teamwork; they receive a hypothetical scenario in which each student is assigned the role of a 'mentor' for a group of younger students who are supposedly embarking on a group project. Their task is to generate five tips/ pieces of advice they would give to these younger students to guide them in collaborating fairly and ethically. In most of the runs of this part of the activity, it was observed that students tended to generate tips mostly drawing on their (negative) past experiences (e.g., 'think carefully before choosing who to work with', 'be prepared to fight with friends').

In the second part, students work in their assigned groups. They are presented with a list of statements/ rules of group work (a list of statements generated by the instructor or derived from literature). Statements refer to a climate of respect, open-mindedness, resolution of conflicts (indicatively: 'All members will respect [other] team members' ideas and promote a positive working attitude', 'If a disagreement should arise, all group members will address the problem in a prompt manner via further email

discussion or in a group meeting’, ‘Each team member will be available for discussion or if help is needed by others’, ‘The team will collectively negotiate expectations to keep the team and the project moving’, ‘Conflicts will be resolved in a respectful manner giving each team member an opportunity to voice their opinion’ (Parker & Goldrick-Jones, 2011, p. 306-307). For this part, each group must discuss and select the three most important statements under the same hypothetical scenario. The Instructor observes group discussions without intervening, keeping notes.

In the third part of the activity, groups cooperate to produce a commonly accepted ‘group code’ by negotiating the terms. The Instructor briefs students on the importance of establishing a common set of rules for their group project. This part requires students to incorporate at least one of the statements suggested by each member in the first part and the statements selected in the second part. Students can also insert other statements that are deemed necessary (for example, many groups of students added statements referring to time management and the need to respect internal deadlines agreed upon by the group).

In the last part of this activity, students reflect on their learning in the form of a class discussion, and they are asked to reflect on their group decision-making processes. They are also encouraged to compare their individual ‘solutions’ to the problem in the first part with the group ‘solutions’ in the third part. The Instructor provides additional feedback, and each group finalizes and signs the contract that will guide subsequent group work.

This activity has received positive comments from students; in the vast majority of cases, it was observed that individuals considered the group's "solution" to be more comprehensive and elaborate than the one they had initially produced, appreciating the merit of their peers’ contributions. It is interesting to note that many students remarked that they liked the idea of agreeing on specific terms for group collaboration.

How could others implement this idea?

Other instructors can supplement the second part of this activity with the introduction of a scoring system to streamline the groups’ decision-making process and/or insert an individual reflective component. In a modified version of this activity, students completed individually a short anonymous questionnaire reflecting on their own contributions after the third part. For the last part of the activity, the Instructor can additionally provide tailored feedback for each group.

Transferability to different contexts

This idea is flexible and adaptable to courses in different fields; it is suitable for skills-building courses. Class sizes need to be small to permit Instructor observation and class discussion in the last part. The activity could also be easily adapted for online courses.

Sources and Attribution

Primary Sources

This section is informed by and adapted from the following sources:

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Use of AI in Section Development

This section was developed using a combination of existing research, expert-informed insights, and AI-assisted drafting. ChatGPT (OpenAI) was used to:

- Synthesize best practices for active learning strategies into a cohesive and accessible guide for instructors.
- Clarify approaches that promote student engagement, participation, and deeper learning.
- Enhance readability and coherence, ensuring that active learning techniques are both research-based and practically applicable in face-to-face and online classrooms.

While AI-assisted drafting provided a structured foundation, all final content was reviewed, revised, and contextualized to ensure accuracy, alignment with research, and pedagogical effectiveness. This section remains grounded in institutional best practices and respects Creative Commons licensing where applicable.

4: [Engaging Minds with Active Learning](#) is shared under a [CC BY-NC-SA 4.0](#) license and was authored, remixed, and/or curated by Antigone Kyrousi.

5: Putting Evidence-Based Instructional Strategies to Work

Teach with evidence, not just enthusiasm.

At a technical college or in any classroom, using evidence-based instructional strategies helps ensure that all students can understand and engage with course material effectively. Two key approaches are Transparency in Learning and Teaching (TILT) and Universal Design for Learning (UDL). TILT focuses on clearly communicating the purpose, tasks, and grading criteria of assignments, which helps students better understand what is expected of them and why it matters. UDL encourages instructors to offer multiple ways for students to access content, demonstrate their knowledge, and stay motivated, allowing for greater flexibility in the learning process. By applying these strategies, instructors can create more structured and supportive learning environments that improve student performance and persistence.

In this section, we will take a closer look at both TILT and UDL, exploring how these evidence-based strategies can strengthen teaching and improve student outcomes in technical college settings. By understanding the core principles of each approach, faculty can begin to identify practical ways to integrate them into their course design and instruction. The goal is to support more effective teaching and learning by making assignments clearer, increasing student engagement, and offering flexible pathways to success. Faculty are encouraged to reflect on their current practices and consider how TILT and UDL can be applied across all courses to enhance the overall learning experience.

Be Transparent

TILT: Transparency in Teaching and Learning

Transparency in teaching means going beyond delivering content; it involves explaining the why and how behind course design and assignments. While instructors often put significant thought into how each activity and assignment can shape students' learning, they rarely make these intentions explicit to students. Transparency asks instructors to openly discuss not just what students are learning but how and why the learning process is structured as it is.

This approach to teaching is especially important for achieving equity in the classroom. Many students, particularly first-generation, low-income, and underrepresented students, may not automatically understand the reasoning behind academic assignments, the optimal ways to tackle them, or the benefits for their learning. These students can spend valuable time trying to decipher the purpose and expectations of an assignment rather than diving straight into the learning itself. By making clear the purpose, specific tasks, and assessment criteria before students start working, instructors provide all students a fairer chance to succeed. Without this clarity, the gap between students with prior knowledge of academic structures and those without can widen, perpetuating the advantages for those more familiar with the “unwritten rules” of academic work.

Some instructors worry that this level of transparency risks simplifying or “dumbing down” the course. In fact, the opposite is true: when students aren't bogged down by trial and error, trying to figure out why an assignment matters or how to approach it, they can focus directly on the essential skills and content of the discipline. This allows students to engage deeply with complex concepts and apply them effectively, ultimately leading to higher-quality work.

Occasionally, a “gatekeeper” mindset suggests that if a student can't intuitively grasp the unstated goals and criteria of an assignment, they aren't suited for the field. But in practice, supporting more diverse ways of thinking benefits not only students but also the discipline. Many academic breakthroughs arise from people who approach problems differently, bringing unique perspectives or interdisciplinary ideas. Transparency in teaching can ensure that more of these “outlier” thinkers—those who see things from a fresh angle—advance to higher levels, contributing valuable insights and fostering innovative research in their fields.

Transparency in teaching and learning can be practically integrated through a variety of strategies that make the expectations and rationale behind assignments and course activities clear to students. Here are some ways transparency can look in practice:

Purpose-Task-Criteria (PTC) Framework:

- **Purpose:** Start each assignment or activity by explicitly explaining its purpose. For instance, “This assignment will help you develop skills in critical reading and synthesis, which are essential for crafting well-supported arguments in your field.”
- **Task:** Describe the specific steps students should take. Avoid general instructions like “analyze the text” and instead clarify steps like “Identify and summarize the main arguments in each article before comparing their approaches.”

- **Criteria:** Share detailed criteria for success, ideally with a rubric or list of specific goals. For example, “A strong paper will clearly explain the main arguments, offer evidence for each point, and consider alternative interpretations.”

Class Activity Rationale:

- Before an activity, explain its purpose and relevance to course goals. For example, “We’re doing this group discussion to practice articulating diverse perspectives, which will help us in analyzing case studies later in the course.”
- After the activity, provide a brief reflection on how the activity ties into larger course concepts and objectives. This helps students connect the immediate task to their broader learning journey.

Assignment Guides and Checklists:

- Create guides for major assignments that outline common pitfalls, best practices, and examples. For example, “An effective research paper in this field typically includes a clear thesis, peer-reviewed sources, and evidence-backed arguments.”
- Provide checklists that students can use to self-assess before submitting. This empowers students to monitor their progress and ensure they meet all assignment criteria independently.

Guided Reflection Prompts:

- Incorporate structured reflection prompts to encourage students to think about how they are learning, not just what they are learning. For instance, after a major assignment, you could ask, “What strategies did you use to approach this assignment? How did those strategies help or hinder your understanding of the topic?”
- Use these reflections as formative feedback to help students identify effective learning strategies, recognize areas for growth, and increase self-awareness.

Real-World Connections:

- Draw connections between assignments and real-world applications, which can be particularly motivating for students. For instance, “Learning how to analyze data trends in this assignment mirrors the kind of work you might do in market research, policy analysis, or environmental science roles.”
- Encourage students to think about how specific skills developed in the course may be useful in their future careers, which helps contextualize their learning.

Collaborative Rubric Development:

- Engage students in co-creating rubrics or criteria for specific projects. This increases investment and helps clarify expectations. For example, ask students what they think an excellent project would include, then work together to refine a rubric based on these ideas.
- Review the rubric together before students begin working, discussing each criterion to ensure students understand what’s expected.

Mid-Semester Check-ins:

- Schedule a “transparency check-in” where students can ask questions about course activities, assignments, and grading criteria. This open dialogue helps address any lingering confusion and gives students insight into the instructor’s goals and teaching approach.
- Encourage feedback on assignments or activities that were particularly helpful or challenging, and adjust future assignments to better meet student needs.

Sharing the “Why” Behind Grading Policies:

- Explain the reasoning behind grading policies, such as why certain assignments are weighted more heavily, why late policies exist, and how each element reflects the learning goals. This can reduce student stress and build trust, as they better understand that policies are designed to support their learning, not just assess it.
- For example, explain, “Your project is weighted more because it integrates skills from the entire semester and is meant to represent a culminating learning experience.”

These practices help create a learning environment where students feel informed, valued, and engaged in their own learning process. By clearly communicating the rationale behind instructional choices, instructors can promote a deeper level of student commitment and enhance the inclusivity of their teaching.

Linked Resources

Visit these resources for more information, templates, handouts, and other freely available materials.

- [TILT Website](http://www.tilthighered.com) [www.tilthighered.com]
- [TILT Transparent Methods](http://www.tilthighered.com/transparent-methods) [www.tilthighered.com/transparent-methods]

Universal Design for Learning (UDL)

Universal Design for Learning (UDL) offers a framework for designing curriculum that provides all students—regardless of their backgrounds, abilities, or needs—with equitable opportunities to learn. Instead of a rigid, one-size-fits-all approach, UDL emphasizes flexibility and customization in instructional goals, methods, materials, and assessments, making learning accessible for everyone.

A Foundation in Variability

UDL is grounded in the understanding that learner variability is not an exception but a predictable part of the educational landscape. This approach acknowledges the full spectrum of diversity in how students engage with, process, and demonstrate their learning. Historically, learners at the “margins” were treated as exceptions, requiring accommodations or special strategies. UDL reframes this perspective, recognizing that designing for variability from the outset benefits all learners.

The Three-Network Model

The UDL framework is based on a three-network model of learning, which considers the brain’s affective (engagement), recognition (representation), and strategic (action and expression) networks. These principles provide a foundation for designing learning environments that are inclusive and adaptable.

- **Engagement:** Offering multiple ways to inspire interest and sustain effort.
- **Representation:** Providing diverse formats for presenting information to ensure accessibility.
- **Action and Expression:** Offering varied methods for learners to demonstrate their understanding.

The Universal Design for Learning Guidelines

The goal of UDL is **learner agency** that is purposeful & reflective, resourceful & authentic, strategic & action-oriented.

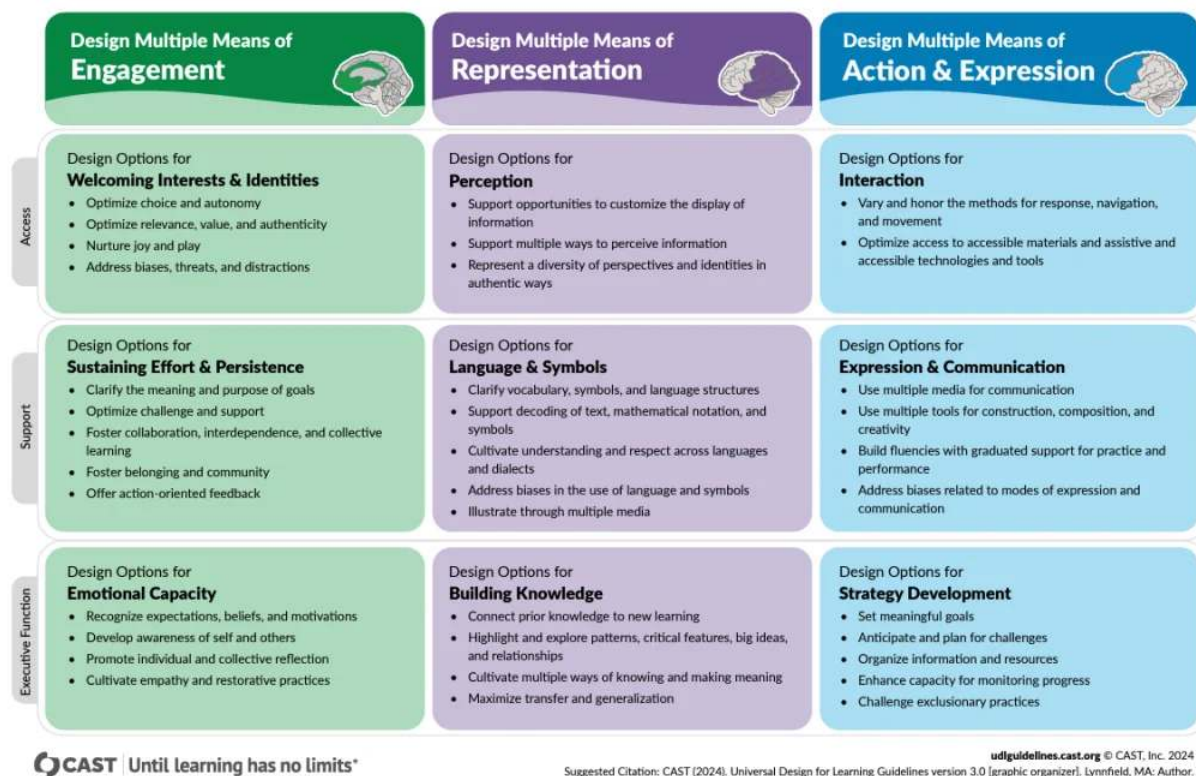


Figure 5.1: UDL 3.0 Framework

Accessible View of Image Text

Guidelines for Application

The UDL Guidelines, developed from an extensive body of research, serve as benchmarks for minimizing barriers in curriculum design. Rather than a fixed checklist or formula, UDL provides principles that are flexible and contextual. These guidelines encourage educators to consider students' developmental stages, cultural contexts, and their own teaching preferences, creating environments that are responsive and dynamic.

UDL as a Translational Framework

UDL acts as a bridge between research and practice. It is not a set of prescriptive tools but a flexible framework for innovation. Educators are encouraged to use UDL principles to design instructional strategies and select resources that fit their specific contexts. For example:

✓ Examples

- Instructors can use digital tools to offer scaffolds and support that adapt to individual needs.
- Low-tech solutions like graphic organizers or tactile materials can be equally effective in addressing learner variability.

The Role of Students

A key aspect of UDL is involving students as active participants in their learning journey. Empowering students to become "experts" in their own learning means fostering self-awareness, persistence, and the ability to articulate their preferences and needs. This collaboration helps create a classroom culture where learning environments are co-designed to enhance access and engagement.

Technology and UDL

While digital technologies can amplify the application of UDL by enabling real-time customization and scaffolded supports, they are not the only solution. Effective UDL implementation blends high-tech and low-tech strategies, leveraging tools and methods that fit the needs of both the learners and the learning context.

Transforming Learning

At its core, UDL is about flexibility—rethinking traditional systems of teaching and learning to prioritize accessibility and inclusion. By designing with variability in mind, educators can create environments where all students thrive, moving beyond accommodation toward a truly inclusive educational experience.

UDL Guidelines 3.0: Expanding Access and Equity

The UDL Guidelines, first introduced in 2008, have been a foundational resource for designing inclusive and accessible learning environments. However, the dynamic nature of education and the evolving understanding of equity and access demand continuous growth. The release of UDL Guidelines 3.0 in July 2024 reflects the culmination of years of research, feedback, and collaborative input from scholars and practitioners around the globe.

This latest iteration explicitly addresses barriers rooted in bias and systemic exclusion, pushing the boundaries of previous versions to honor the complexity of learner variability and emphasize the importance of identity in the learning process. By integrating asset-based approaches, UDL 3.0 aligns itself more closely with broader movements in culturally responsive teaching and equity-focused pedagogy.

Key Expansions in UDL Guidelines 3.0

- 1. Integrating Asset-Based Frameworks:** UDL 3.0 engages in dialogue with other equity-centered and culturally sustaining pedagogies, emphasizing their complementary goals. This approach acknowledges and values the cultural and linguistic practices learners bring into the classroom, creating a richer, more inclusive framework for learning design.
- 2. Recognizing Identity as Part of Variability:** While previous UDL iterations focused on how learners engage, perceive, and express learning (the “why,” “what,” and “how”), the updated guidelines explicitly incorporate identity as the “who” of learning. By recognizing students’ intersecting identities—including race, culture, language, gender, and ability—UDL 3.0 encourages educators to design environments that affirm and celebrate this diversity.
- 3. Addressing Bias at All Levels:** The updated guidelines emphasize the need to confront individual, institutional, and systemic biases as fundamental barriers to learning. By doing so, UDL 3.0 aims to dismantle exclusionary practices that limit opportunities for certain groups and foster more equitable access to education.
- 4. Valuing Interdependence and Collective Learning:** Moving beyond the traditional emphasis on individual achievement, the new guidelines highlight the importance of collaboration and shared growth. This shift acknowledges the value of community and collective knowledge-building in fostering deeper, more meaningful learning experiences.
- 5. Adopting Learner-Centered Language:** A significant shift in UDL 3.0 is the use of language that reflects a learner-centered perspective. Verbs and phrases now apply interchangeably to educators and students, reinforcing the idea of shared responsibility and agency in designing and navigating learning experiences.

Enhanced Themes Within UDL Principles

The three core UDL principles—Engagement, Representation, and Action and Expression—are expanded in Guidelines 3.0 to deepen their connection to equity and inclusion:

Engagement:

- Affirming learners’ identities and interests
- Promoting belonging as a foundation for learning
- Emphasizing joy, play, and restorative practices in education

Representation:

- Showcasing authentic diversity in identities and narratives
- Valuing multiple cultural perspectives and ways of knowing

- Addressing historical erasure and bias in curricula

Action and Expression:

- Honoring diverse modes of communication, including those historically marginalized
- Challenging exclusionary practices to create equitable systems
- Providing space for creativity and self-expression that reflects individual and cultural authenticity

Connecting UDL 3.0 to Broader Access and Equity Efforts

The updates in UDL Guidelines 3.0 resonate strongly with themes of access and equity woven throughout this guide. By emphasizing the systemic and cultural dimensions of inclusion, UDL 3.0 offers educators a powerful lens through which to evaluate and refine their teaching practices. Its focus on flexibility, identity, and interdependence aligns with a shift from a deficit-based to an asset-based understanding of learner variability.

By incorporating UDL 3.0 principles, educators can design environments where all learners—especially those historically excluded or marginalized—can thrive. This framework invites us to re-imagine teaching and learning as a collaborative, dynamic, and deeply human process that values the richness of every learner’s identity.

Discussion Questions

1. In what ways might clearly communicating learning goals alongside offering multiple ways to engage and express understanding help reduce student confusion and increase confidence?
2. How might pairing clear, transparent assignment instructions with varied learning materials and assessment options support students in taking ownership of their learning?
3. How can combining TILT’s focus on transparency with UDL’s flexible approaches reduce barriers and encourage all students to demonstrate their true abilities?

Sources and Attribution

Primary Sources

This section is informed by and adapted from the following sources:

- Transparency in Learning and Teaching (TILT) Higher Ed Project. TILT Higher Ed Resources and Research. Available at: [TILT Higher Ed Website \(opens in new window\)](#)
- Project Information Literacy (2022). Smart Talk Interview: Transparency in Teaching and Learning. Available at: [Project Information Literacy \(opens in new window\)](#)
- CAST. Universal Design for Learning (UDL) Guidelines. Available at: [UDL Guidelines \(opens in new window\)](#)
- CAST. UDL on Campus: Universal Design for Learning in Higher Education. Available at: [UDL on Campus \(opens in new window\)](#)

Use of AI in Section Development

This section was developed using a combination of existing research, expert perspectives, and AI-assisted drafting. ChatGPT (OpenAI) was used to:

- Organize and refine key concepts from TILT research into a cohesive and accessible framework for instructors.
- Clarify practical applications of transparency strategies in teaching and learning.
- Ensure readability and engagement while preserving the core principles of evidence-based transparency practices.

While AI-assisted drafting provided a structured foundation, all final content was reviewed, revised, and contextualized to maintain accuracy, alignment with research, and practical applicability. This section remains grounded in scholarly and institutional best practices and respects Creative Commons licensing where applicable.

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6: Navigating AI in Teaching and Learning

AI generated this quote. I just spent 20 minutes reading 100 variations to decide if it was the right kind of clever.

Adapting Teaching Strategies to Rapid AI Advancements

The pace of change in artificial intelligence (AI) is unlike anything seen in recent educational technology. In technical college settings, where students are being prepared for fast-evolving industries, it's crucial that teaching strategies reflect the dynamic nature of AI. Tools like ChatGPT, Copilot, Claude, Magic School, and Diffit are becoming increasingly embedded in how students learn, collaborate, and solve problems. These AI-powered platforms offer personalized learning support, automate complex tasks, and even assist in coding or content generation, transforming both the teaching and learning experience.

Due to the rapid development of these technologies, it's important for instructors to understand that the resources and tools shared in a given semester may become outdated by the next. For example, features in ChatGPT or Claude might expand significantly within a few months, or new tools like Magic School and Diffit might introduce capabilities that reshape instructional design. This constant evolution necessitates a flexible and adaptive approach to curriculum planning, with a focus on teaching students how to evaluate and effectively utilize emerging tools rather than relying solely on fixed content.

Faculty play a crucial role not only in incorporating AI tools into their teaching but also in guiding students to use them responsibly and effectively. It is important that instructors themselves engage with platforms like ChatGPT, Copilot, and Claude in order to understand both their strengths and limitations. By doing so, educators can better mentor students to use AI as a tool for learning and growth—encouraging them to revise, analyze, and refine the AI's output rather than accepting it at face value. This helps foster critical thinking and digital literacy skills, which are essential in both academic and professional contexts.

One practical implication is that course materials, assignments, and examples need to be reviewed and updated regularly. AI tools such as Copilot, which assist with coding by predicting and suggesting code, can significantly alter how programming is taught and practiced. Similarly, Magic School and Diffit provide AI-powered lesson planning and differentiation support, which may shift instructional methods in education and training fields. As educators, staying informed through professional development and industry updates is critical to maintaining the relevance and accuracy of instructional content. Ultimately, teaching students how to learn with AI—not just use it—will prepare them to succeed in an ever-changing technological landscape.

Guiding Students in Responsible AI Use

What we know about AI in the classroom is that students are using it—whether we want them to or not. It is up to us as faculty to give guidance on how to navigate this evolving landscape in ways that support learning rather than undermine it.

This includes:

1. helping students understand how to use AI tools to deepen their learning and critical thinking
2. clearly defining the expectations and boundaries for AI use in each assignment or lesson, and
3. connecting AI use to real-world applications in their chosen industry, so they can develop the skills and judgment needed to use these tools effectively and responsibly in their future careers.

Ensuring Learning

Using AI means guiding students to see these tools as support, not substitutes, for their own thinking and effort. Faculty can help students use AI to brainstorm ideas, clarify complex topics, generate practice problems, or receive feedback on drafts, all while emphasizing the importance of critical evaluation. Rather than simply copying AI-generated content, students should be encouraged to question, refine, and build upon it. This approach helps them stay actively engaged in the learning process and develop essential skills like analysis, synthesis, and problem-solving. By modeling appropriate uses of AI and integrating it into assignments thoughtfully, instructors can help students use these tools to enhance their education.

Students should know the AI expectations for any assignment or course from the start. Providing clear guidelines (such as the examples below) helps students understand when, how, and if AI tools can be used. Just as important as the guidelines themselves is explaining the why behind them. For example, telling students, “I want you to brainstorm with a partner and not be influenced by

AI at this point,” communicates that the goal is for them to draw on their own ideas and prior knowledge first. When students understand the purpose behind the parameters, whether it’s to build foundational thinking, practice a specific skill, or simulate industry expectations, they are far more likely to respect and follow them. On the other hand, if a course is simply labeled “AI is not permitted,” without further context, many students may disregard the restriction and use AI anyway. Transparency, consistency, and explanation go a long way in shaping responsible and intentional use of AI in the classroom.

Below is a very simple version of an AI Usage Guide – simply labeling each of your assessments with a red, yellow, or green light, and then giving guidance on what that would mean for students, can be an easy way to find balance and give expectations.

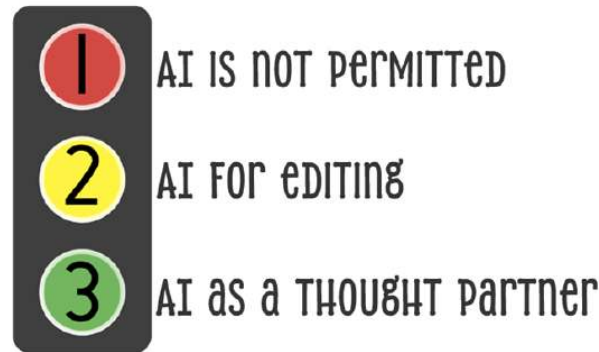


Figure 6.1: AI label for assessments.

This second AI Usage Guide is more comprehensive. The expectations are clearly provided, and students should be able to understand how, when, and why they should or should not be using AI.

AI Usage Guide for Assignments				
Level One	Level Two	Level Three	Level Four	Level Five
In Your Own Words	AI as Limited Reference Use	Process Assistance Permitted with AI	AI Guided Content Collaborator	AI as a Co-Creator
Defined	Defined	Defined	Defined	Defined
All work completed without AI assistance	AI may be consulted like a dictionary or encyclopedia	AI may be used as a process tool but not for producing the final content	AI contributes directly for content creation with student direction and evaluation	Full AI integration in the creative and academic process
For Example,	For Example,	For Example,	For Example,	For Example,
Traditional research methods, tutors, library resources	Asking AI factual questions, requesting explanation of concepts	Brainstorming, outlining, proofreading	Generation of draft documents, suggested approach to problem solve, feedback on work	AI generated content, AI developed ideas, AI analysis and interpretation
Rationale	Rationale	Rationale	Rationale	Rationale
Develops fundamental skills without technological shortcuts; ensures authentic demonstration of core competencies.	Allows AI as a learning resource while maintaining student authorship of all submitted work.	Acknowledges AI as a productivity tool while ensuring critical thinking and composition remains student driven.	Teaches critical engagement with AI outputs and develops skills in directing and improving machine-generated content.	Prepares students for a future where AI collaboration is normalized while developing meta-skills in effective human-AI teamwork.

Figure 6.2: AI Usage Guide for Assignments

[Accessible view of image text](#)

AI Usage Guidelines for Assignments

This guide provides a framework for faculty to clearly communicate expectations regarding AI tool usage in academic assignments. The framework consists of five progressive levels, from complete prohibition to full integration of AI as a collaborator.

How to Use: Faculty will assign each assessment a level, so the expectation of AI usage is understood between faculty and students.

Level 1: No AI Use Permitted

- **Definition:** All work must be completed without any AI assistance.
- **Acceptable:** Traditional research methods, library resources, approved software tools, consultation with tutors or writing centers.
- **Prohibited:** Any use of generative AI tools, including for brainstorming, drafting, editing, or checking work.
- **Rationale:** Develops fundamental skills without technological shortcuts; ensures authentic demonstration of core competencies.
- **Example Statement:** "This assignment must be completed entirely without AI assistance. All work should represent your own thinking, writing, and problem-solving abilities."

Level 2: Limited Reference Use Only

- **Definition:** AI may be consulted for reference information only, similar to using an encyclopedia.
- **Acceptable:** Asking AI factual questions, requesting explanations of concepts, and generating examples for learning purposes.
- **Prohibited:** Using AI to produce any content that appears in the final submission, having AI critique or improve your work.
- **Optional Documentation Required:** Citations for any information obtained from AI systems.
- **Rationale:** Allows AI as a learning resource while maintaining student authorship of all submitted work.
- **Example Statement:** "You may use AI tools as reference sources only. Any information obtained must be cited, and all writing and analysis must be entirely your own."

Level 3: Process Assistance Permitted

- **Definition:** AI may be used as a process tool, but not for producing final content.
- **Acceptable:** Brainstorming ideas, outlining, generating research questions, checking grammar/spelling, and simplifying complex texts for better understanding.
- **Prohibited:** Submitting AI-generated text, code, or solutions as your own work.
- **Optional Documentation Required:** A detailed description of how AI was used in your process.
- **Rationale:** Acknowledges AI as a productivity tool while ensuring critical thinking and composition remain student-driven.
- **Example Statement:** "You may use AI tools to assist your process (brainstorming, outlining, editing), but all final content must be your original work. Include a paragraph describing how you used AI in your process."

Level 4: Guided Content Collaboration

- **Definition:** AI may contribute directly to content creation under student direction and with critical evaluation.
- **Acceptable:** Using AI to generate draft sections, create visualizations, suggest approaches to problems, and provide feedback on work.
- **Requirements:** Substantial revision and improvement of AI outputs, critical evaluation of AI suggestions, and clear student ownership of the final work.
- **Optional Documentation Required:** Detailed documentation of all AI inputs and student modifications.
- **Rationale:** Teaches critical engagement with AI outputs and develops skills in directing and improving machine-generated content.
- **Example Statement:** "You may collaborate with AI to generate content, but you must significantly revise all AI outputs, critically evaluate all suggestions, and document the collaboration process. Final submissions should demonstrate your judgment and expertise in directing the AI and refining its work."

Level 5: AI as Co-Creator

- **Definition:** Full integration of AI as a collaborative partner in the creative/academic process.
- **Acceptable:** Extensive AI generation of content, code, or solutions; ongoing dialogue with AI to develop ideas; AI-assisted analysis and interpretation.
- **Requirements:** The assignment focus shifts to how effectively students can leverage AI capabilities, provide appropriate prompts, evaluate outputs, and integrate AI work with human creativity and judgment.

- **Optional Documentation Required:** Comprehensive documentation of the collaboration, including prompts used, decision-making process, and reflective analysis of the collaboration.
- **Rationale:** Prepares students for a future where AI collaboration is normalized while developing meta-skills in effective human-AI teamwork.
- **Example Statement:** "For this assignment, you will work with AI as a co-creator. Your grade will be based not on avoiding AI assistance but on how skillfully you direct the AI, integrate its contributions, and reflect critically on the collaboration process. Include all prompts used and a detailed reflection on your collaborative process."

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7: Discussion Bored or Discussion Board?

I promise you no one agrees more than college students on a discussion board.

Discussion Bored or Discussion Board?

Discussion boards have long been a staple of online and hybrid learning, but they often fall flat when prompts are too generic or disconnected from students' interests. To make discussion boards more engaging and meaningful, whether in an online course or as a supplement to in-person learning, consider using them in fresh, creative ways. For example, instead of traditional question-and-answer threads, try using role-play scenarios where students take on perspectives from their future industry roles to solve a problem. This not only makes the discussion more dynamic but also helps students apply course content in context.

Rather than using discussion boards strictly for assessment, consider designing them to mimic the feel of small group discussions - spaces where students can play with ideas, share experiences, and reflect without the pressure of being "graded." Encourage informal, exploratory conversation where students can bounce thoughts off one another, ask genuine questions, and post imperfect or in-progress thinking. This approach fosters a sense of community and curiosity, helping students feel more connected and engaged. When the focus shifts from evaluation to interaction, discussion boards become places for authentic learning and creativity rather than routine compliance.

Rethinking the Traditional Discussion Board Model

Key Takeway

Critique: The common "post and respond to two classmates" model can feel formulaic and discourage authentic dialogue.

Shift: Create student-centered discussion boards that emphasize conversation, critical analysis, debate, and creativity.

Examples of Re-imagined Discussion Board Activities

Discussion Board as Annotation Space

Activity: Students use the discussion board to collaboratively annotate a text, image, or video.

Example: In a film studies course, students discuss camera angles, symbolism, and narrative techniques by annotating stills or clips.

Discussion Board as Presentation Space

Activity: Students post presentations with video, audio, or visual aids.

Setup: Students upload multimedia presentations, and peers provide feedback based on a rubric.

Example: In a marketing course, students present product pitches and receive constructive feedback from their classmates.

Discussion Board as Gallery Space

Activity: Use visual content to enhance discussion and stimulate creativity.

Example: In a literature course, students create memes based on themes from assigned readings and analyze how humor or imagery conveys the text's messages.

Discussion Board as Work Space

Activity: Use small-group boards for collaborative problem-solving or project work.

Benefits: Boards provide a record of group progress and can facilitate a modified think-pair-share activity.

Example: In a coding class, groups debug provided code snippets and document their process and solutions for peer review.

By integrating external tools, AI-supported learning, and ethical considerations, instructors can create vibrant online learning environments. These approaches encourage active participation, foster deeper connections to the material, and build essential skills like collaboration, critical thinking, and creativity.

Discussion Questions

1. Which of the re-imagined discussion board formats do you think would be most effective in your subject area or course? Why?
2. How can rethinking the discussion board from a grading tool to a creative or collaborative space change student engagement and ownership of learning?
3. What challenges might arise when implementing these alternative discussion board formats, and how can instructors support students in navigating them successfully?

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8: Supporting Students to Self-Advocacy

Building time management skills today builds self-advocates for tomorrow.

Key Strategies for Supporting Time Management and Self-Regulation

Time management and self-regulation are crucial skills for academic success, yet many students struggle with balancing competing demands. Instructors can play a pivotal role in supporting the development of these abilities by creating clear structures, offering resources, and encouraging regular reflection. When instructors integrate transparency, provide practical tools, and build in opportunities for feedback and accountability, they not only help students manage their time more effectively—they also empower them to take ownership of their learning. Over time, this support can evolve into self-advocacy, as students gain the confidence and skills to identify what they need, seek out resources, and make informed decisions that support their academic and personal goals.

Emphasize Transparency in Task Expectations

Building on the [TILT \(Transparency in Learning and Teaching\)](#) framework, instructors can design assignments and activities that are clear and accessible to all students.

Clearly Define Tasks

- Provide detailed instructions that outline what students are expected to do, how they will be assessed, and why the task is important.
- **Example:** Instead of assigning “Write a paper on X,” specify: “Write a 3-5 page analytical paper discussing X, using at least three peer-reviewed sources. Submit a draft for feedback by [date].”

Estimate Time Requirements

- Include realistic time estimates for each task to help students plan their schedules.
- **Example:** “Reading this article should take approximately 30-45 minutes, and writing your reflection will take another 20-30 minutes.”

Provide Models and Examples

- Share examples of successful work to give students a clearer understanding of expectations.

Embed Feedback Opportunities

Opportunities for feedback help students gauge their progress and adjust their strategies.

Scaffold Assignments

- Break larger tasks into smaller components with deadlines and feedback at each stage.
- **Example:** A research project could include a topic proposal, annotated bibliography, draft, and final submission, each with time for instructor or peer feedback.

Offer Self-Check Prompts

- Include questions or rubrics for students to evaluate their own work before submission.
- **Example:** “Have I addressed all parts of the prompt? Is my thesis clear? Have I cited sources correctly?”

Normalize Iteration

- Encourage students to view feedback as part of the learning process rather than a final judgment.

Share Time Management Tools and Resources

Providing students with practical tools can empower them to organize their time effectively.

Introduce Planning Tools

- Share templates for time management, such as weekly planners, study schedules, or online tools like Google Calendar, Trello, or Notion.
- **Example:** Offer a downloadable spreadsheet with columns for tasks, deadlines, estimated time, and progress tracking.

Promote Use of Apps and Software

- Recommend apps like [Pomofocus \(opens in new window\)](#) (Pomodoro technique), [Forest \(opens in new window\)](#) (time tracking), or [MyStudyLife \(opens in new window\)](#) (task organization).

Integrate Time Audits

- Encourage students to track their time for a week to understand how they allocate it and identify areas for improvement.
- **Example:** Ask students to record how much time they spend on studying, attending class, and leisure, then reflect on the balance.

Build Reflection into Course Activities

Encouraging students to reflect on their time management habits can improve their self-awareness and regulation.

Include Time Reflection Prompts

- After a major assignment or project, ask students to reflect on how they planned and managed their time.
- **Example Questions:**
 - “How much time did you dedicate to this task? Was it sufficient?”
 - “What strategies worked well for managing your time, and what would you change?”

Use Mid-Semester Check-Ins

- Include a reflection activity midway through the term to help students evaluate their time management and make adjustments.
- **Example:** “What is one thing you could start, stop, or continue doing to improve your study habits?”

Incorporate Metacognitive Wrappers

- Combine reflection with learning by using exams or assignment wrappers.
- **Example:** “Before you begin studying for the final, outline how you plan to prepare and how you will track your progress.”

Foster a Supportive Environment

Help students feel supported in their efforts to manage time and self-regulate.

Acknowledge Challenges

- Validate that time management is a skill to be learned, not an inherent trait.
- **Example:** Share stories of your own struggles with time management and how you overcame them.

Encourage Peer Support

- Create opportunities for students to share tips and strategies with one another, such as in discussion boards or study groups.

Provide Flexibility When Possible

- Recognize that students’ circumstances vary and offer reasonable flexibility with deadlines or extensions when appropriate.

Bringing It All Together: Supporting Lifelong Skills

Instructors who intentionally build transparency, regular feedback, practical tools, and opportunities for reflection into their courses play a vital role in helping students develop important time management and self-regulation skills. These skills are foundational not only for academic success but also for managing life’s various demands beyond the classroom. When students understand expectations clearly and receive consistent guidance, they are better equipped to organize their time, monitor their progress, and adjust their strategies as needed.

Beyond improving grades, these teaching practices cultivate a growth-oriented mindset that encourages students to view challenges as opportunities for learning and development. By modeling effective strategies and supporting reflection, instructors empower

students to take ownership of their education. This sense of agency prepares students to navigate future personal and professional situations with confidence, making these skills essential for success both inside and outside of school.

AI Attribution Statement

This section on Helping Students Self-Regulate and Manage Their Time was developed using AI-assisted drafting based on previously generated content from this guide.

Use of AI in Section Development

- ChatGPT (OpenAI) was used to synthesize and refine strategies for time management, self-regulation, and academic planning, ensuring consistency with research-based best practices already outlined in this guide.
- The AI-assisted process included restructuring and clarifying key insights, making the content more cohesive and practical for educators and students.
- While AI-assisted drafting provided organizational support and refinement, all final content was reviewed, revised, and aligned with the broader themes of self-directed learning and metacognitive skill development.

This section remains grounded in research-based teaching strategies and aligns with best practices in supporting student success.

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